AI MENTAL HEALTH UK

Task - Distribution Analysis

Duration – 6-8 May

Name – Rushay Gopani

Domain - Data Science

DS-3 Report

Distribution Analysis using APMS 2014

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1. Introduction

This analysis was conducted using the Adult Psychiatric Morbidity Survey (APMS) 2014 dataset published by NHS Digital and NatCen Social Research. The study focused on analysing the distribution of Common Mental Disorder (CMD) severity scores across different age groups.

The analysis also examined whether the reconstructed mental health severity distribution approximately followed a normal distribution using histogram and QQ plot analysis.

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2. Objective

The objectives of DS-3 were:

- **To analyse CMD severity score distributions.**
- **To reconstruct population-level severity distributions.**
- **To visualise prevalence distributions using histograms.**
- **To evaluate normality using QQ plots.**
- **To identify skewness and uneven distribution behaviour.**

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3. Dataset Used

The analysis used APMS 2014 data from:

Table	Purpose
Table 2.1	CIS-R severity distribution
Table 2.3	Anxiety and depression prevalence
Table 2.11	Regional CMD prevalence

The main analysis was based on Table 2.1.

	Age_Group	0_5	6_11	12_17	18_plus
0	16-24	67.1	15.5	7.8	9.5
1	25-34	66.3	16.5	9.0	8.2
2	35-44	66.4	15.7	8.1	9.8
3	45-54	65.3	16.7	8.2	9.8
4	55-64	66.3	16.7	7.8	9.2
5	65-74	76.6	13.2	5.9	4.2
6	75+	77.6	14.3	4.7	3.3

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4. Methodology

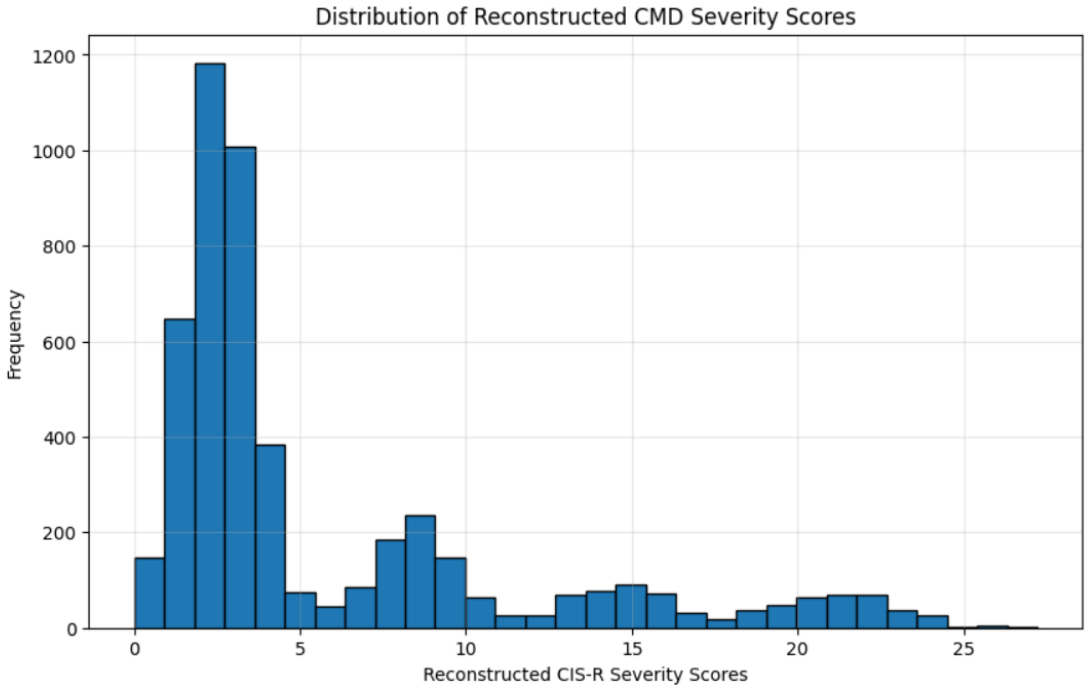
The following steps were performed:

1. **Extracted CIS-R severity band prevalence values.**
2. **Reconstructed pseudo-individual severity scores using prevalence-weighted sampling.**
3. **Generated histogram distributions.**
4. **Applied QQ plot analysis to assess approximate normality.**
5. **Interpreted skewness and prevalence patterns.**

The severity bands used were:

- **0–5 → minimal symptoms**
- **6–11 → mild symptoms**
- **12–17 → moderate symptoms**

- **18+ → severe symptoms**

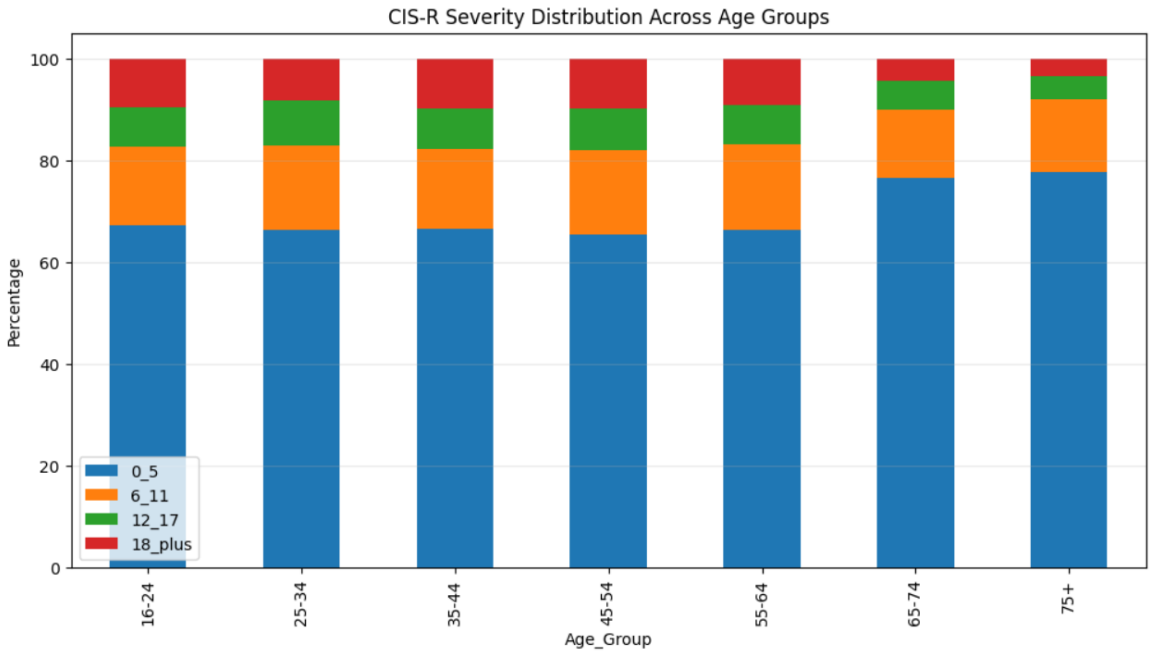


5. Histogram Analysis

Histogram analysis showed that CMD severity scores were not evenly distributed across the population.

Most individuals were concentrated within lower severity score ranges, while a smaller proportion exhibited moderate-to-severe psychological distress.

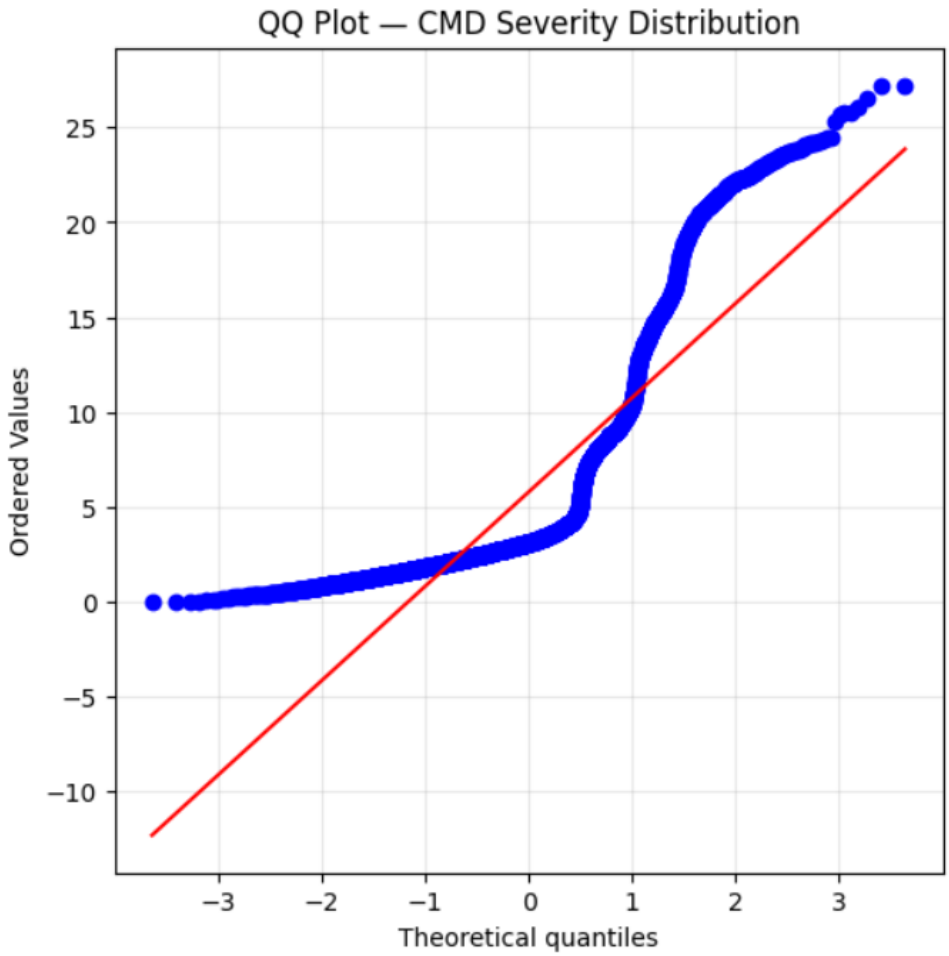
This created a mildly right-skewed distribution.



6. QQ Plot Analysis

The QQ plot showed that most points approximately followed the reference line, indicating approximate normality.

However, deviations at the distribution tails suggested mild skewness and uneven prevalence distribution across demographic groups.



7. Key Findings

The analysis identified several important patterns:

- **Most adults showed lower CMD severity levels.**
- **A smaller high-risk population exhibited higher psychological distress.**
- **The severity distribution was mildly right-skewed.**
- **QQ plots indicated approximate normality with slight deviations.**

8. Conclusion

The DS-3 distribution analysis successfully modelled CMD severity score distributions using real APMS 2014 data.

The findings showed that mental health severity is unevenly distributed across the population, with most individuals experiencing lower symptom levels and smaller groups showing elevated psychological distress.

Histogram and QQ plot analysis demonstrated approximate normality with mild skewness in the reconstructed severity distribution.

9. References

1. **NHS Digital — APMS 2014**
2. **NatCen Social Research**
3. **APMS Chapter 2 Tables**
4. **Python, Pandas, NumPy, Matplotlib, SciPy**